AIRVO™2 and AIRVO™3 Standard Operating Procedure

Paediatric Intensive Care Unit (PICU)
Cardiology (Ward 1E)
Medical Acute Receiving Hi-Flow in Bronchiolitis

Royal Hospital for Children Glasgow

1. Introduction

High Flow Nasal Cannula (HFNC) Therapy delivers warmed and humidified respiratory gases to spontaneously breathing patients. It is generally well tolerated and provides a level of positive airway pressure similar to nasal Continuous Positive Airway Pressure (nCPAP). Compared with nCPAP, HFNC therapy is associated with less nasal trauma.

HFNC therapy has a strong safety profile. However, abdominal distension may occur and should be monitored throughout treatment. Studies have demonstrated that HFNC therapy can reduce the need for intubation in infants and children with acute respiratory failure. The system delivers flow rates that meet or exceed the patient's inspiratory demand, reducing dilution of oxygen therapy.

Within the Royal Hospital for Children Glasgow (RHC) AIRVO™2 is the primary device used to deliver HFNC therapy. AIRVO™3 is used in specific clinical circumstances (see AIRVO™3 Patient Pathway). Both devices are manufactured by Fisher & Paykel Healthcare and are licensed for patients greater than 3kg, therefore for patients under this weight, treatment will be in the Paediatric Intensive Care Unit utilising the MR850 device – please refer to Humidified High Flow Nasal Cannulae (HHFNC) | NHSGGC for any patients under 3kg.

All AIRVO™ devices improve patient comfort and support secretion clearance. Oxygen concentrations and flow can be adjusted according to clinical need for individual patients.

2. Objectives

The objective of this SOP is to provide health professionals with guidance for a consistent approach to delivering HFNC therapy via either Airvo™2 or Airvo™3, promoting a standardised approach, ensuring increased patient safety and continuity of patient care.

3. Scope

This SOP applies to staff caring for patients receiving AIRVO™2 or AIRVO™3 therapy within RHC, including:

- PICU (Ward 1D)
- Cardiology Ward (Ward 1E)
- Acute Receiving Unit (Ward 2C)

Staff should follow the guideline relevant to their clinical area.

4. Limitations

Please note this protocol is not applicable to staff caring for children supported in hospital/community settings on a MY AIRVO™ device, with patient specific prescriptions. For help and support related to this system please contact the child's specialist team.

PICU HFNC Initiation Guide

Hi-Flo Initiation Guide

Indications for HiFlo:

- ✓ Moderate-severe bronchiolitis
- ✓ Cardiac failure – mild
- ✓ Respiratory failure
  - ○ Following extubation in PICU
  - ○ Pneumonia, pulmonary oedema, atelectasis
- ✓ Moderate asthma
- ✓ Failure to thrive "blow & grow" strategy

Contra-indications to HiFlo:

- ❖ Multi-organ failure
- ❖ Trauma / Surgery to nasopharynx & base of skull
- ❖ Congenital abnormality of nasopharynx eg choanal atresia

Monitoring

- Continuous cardiac and saturation monitoring
- Hourly PAWS/CEWS (if used in clinical area)
- Hourly Hi-Flo recordings
- Minimum 4 hourly blood pressure
- Minimum 4 hourly medical review

Hi-Flo initiation	Neonate <3kg	Neonate >3kg	1-12 months	1-4yrs	5-10yrs	10yrs+
HiFlo device	MR850	AIRVO	AIRVO	AIRVO	AIRVO	AIRVO
Cannula Size (<50% nare) (EST.)	Neonate *(yellow)*	Infant *(Purple)*	Infant *(Purple)*	Paediatric *(Green)*	Paediatric *(Green)*	S / M / L *(Blue)*
Humidifier Temp.	34°C	34°C	34°C	34°C	34°C	37°C
Initial Settings	8L/min FiO_2 40% (Unless cardiac with balanced circulation)	10L/min & adjust O2 flow to give FiO_2 40% (Unless cardiac with balanced circulation)	12L/min, & adjust O2 flow to give FiO_2 40% (Unless cardiac with balanced circulation)	15L/min & adjust O2 flow to give FiO_2 40% (Unless cardiac with balanced circulation)	20L/min & adjust O2 flow to give FiO_2 40% (Unless cardiac with balanced circulation)	30L/min & adjust O2 flow to give FiO_2 40% (Unless cardiac with balanced circulation)
Max Flow Rate	8L/min	20L/min	20L/min	25L/min	25L/min	50 / 60 / 60
Escalation criteria • Oxygen saturations (SpO2) <94% (NB in cyanotic congenital heart disease this may not apply) • Tachypnoeic, persistent or worsening respiratory effort despite HiFlo +/- increased WOB • Tachycardia, persistant or worsening despite HiFlo • Sustained elevated respiratory component of PEWS score						
1st Escalation	Titrate FiO_2 to SpO_2≥94% Unless cardiac with balanced circulation	Titrate FiO_2 to SpO_2≥94% Unless cardiac with balanced circulation	Titrate FiO_2 to SpO_2 ≥94% Unless cardiac with balanced circulation	Titrate FiO_2 to SpO_2 ≥94% Unless cardiac with balanced circulation	Titrate FiO_2 to SpO_2 ≥94% Unless cardiac with balanced circulation	Titrate FiO_2 to SpO_2 ≥94% Unless cardiac with balanced circulation
2nd Escalation	Increase flow up to 8 & urgent PICU review	Increase flow up to 20 & urgent PICU review	Increase flow up to 20 & urgent PICU review	Increase flow up to 25 & urgent PICU review	Increase flow up to 25 & urgent PICU review	Increase flow up to max & urgent PICU review
3rd Escalation	nCPAP/Intubate	nCPAP/Intubate	NIV/Intubate	NIV/Intubate	NIV/Intubate	NIV/Intubate

HiFlo for Apneoic oxygenation peri-intubation	Neonate (up to 1 month)	1-12 months	1-4yrs	5-10yrs	10yrs+
Cannula Size (<50% nare)	Infant	Infant	Paediatric	Paediatric	S / M / L
Humidifier Temp. (°C)	34	34	34	34	34
Set Flow Rate (L/min)	2L/kg/min	2L/kg/min	25	25	50 / 60 / 60
FiO_2 (%) *Unless cardiac with balanced circulation*	100	100	100	100	100

***Please note**, this guide includes reference to the MR850 High Flow device. This device is NOT available for use throughout the rest of the hospital. Please refer to for MR850 guidance [Humidified High Flow Nasal Cannulae (HHFNC) | NHSGGC]

Hi-Flo Weaning Guide

Is the child clinically stable for nurse-led weaning of Hi-Flo therapy?

Ensure:

- ✓ Observations stable or within normal limits:
  - Heart rate, respiratory rate, saturations, temperature and blood pressure
- ✓ No evidence of increased work of breathing:
  - No subcostal, intercostal or sternal recession
  - No tracheal tug
  - Absence of nasal flaring, head bobbing and lethargy
- ✓ Non-cyanotic cardiac patient
- ✓ Latest blood gas stable (if taken)
- ✓ Circulation stable
- ✓ Agree with Medical staff patient is suitable for nurse-led Hi-Flo weaning pathway

YES

NO

Discuss with Medical Staff
- Agree escalation plan
  - nCPAP or Intubation
- Agree Medical weaning plan

Monitoring
- Continuous cardiac and saturation monitoring
- Hourly PAWS/CEWS (if used in clinical area)
- Hourly Hi-Flo recordings
- Minimum 4 hourly blood pressure
- Minimum 4 hourly medical review

Wean Plan (step 1 – wean oxygen then wean flow
- Wean Oxygen concentration by 5% every hour to 40%
  - Review patient clinically & evaluate changes with each wean of oxygen, Target SaO2 ≥94%
- Once Oxygen of 40% achieved wean flow by 2 Litres per minute (LPM) every 2 Hours until a flow of 4 LPM is achieved
  - Review patient clinically & evaluate changes with each wean of flow, Target SaO2 ≥94%

Wean Plan (step 2 – convert to nasal cannula oxygen)
- Once oxygen of 40% and Flow of 4 LPM is achieved and patient stable for 2 hours convert to nasal cannula oxygen at 3 LPM, Target SaO2 ≥94%
- Review patient clinically & evaluate changes with each wean of flow, Target SaO2 ≥94%

Remember:
With any deterioration in the patients condition reverse previous step, re-evaluate and inform medical staff.

Version 4.1
September 2017

Cardiac Ward guide for initiating and managing patients on AIRVO™

NHS Greater Glasgow and Clyde

Cardiac Ward guide for Hi-Flo - initiating and managing patients on Airvo

INDICATIONS:

- Clinical diagnosis of cardiac pathology who is ≥3kg
- Increase work of breathing ,tachycardia
- Faltering growth 'blow and grow'
- PICU review & Cardiac reg/ANP/H@N joint agreement for the need for the initiation of Hi-Flo
- Consultant cardiologist aware and in agreement
- Ward 1e staffed to accommodate HiFlo patient

CONTRAINDICATIONS for Ward 1e HiFLo:

- Weight <3kg unless agreed with consultant and senior charge nurse
- Confirmed diagnosis of bronchiolitis (these patients are looked after in Ward 2c or PICU)
- Congenital abnormality of nasopharynx e.g. choanal atresia

MONITORING:

- Continuous ECG and saturation monitoring, once stable and established can remove continuous ECG
- Hourly PEWS until stable on Hi-Flo then can go to 2hrly PEWS.
- Minimum 4 hourly blood pressure
- Flag as a "HDU level" at Huddle
- Ensure appropriate PEWS targets are agreed, set and documented

MOVING & STAFFING:

- Airvo2 with battery pack should be used to move patient (available from PICU)
- Staff at HDU level (ie 1 nurse : 2 patients)

Hi-Flo initiation	3kg - 8kg	>8kg – 25kg	>25kg
Hi-Flo device	AIRVO 2	AIRVO 2	AIRVO 2
Tubing size	Junior	Junior	Adult
Cannula size (to cover <50% nare)	Infant (Purple)	Paediatric (Green)	Adult (Blue) Small or Medium
Humidifier temperature	34°C	34°C	34°C
Settings	Max 2 L/kg/min & adjust O2 flow to target appropriate SpO2	Max 2 L/kg/min & adjust O2 flow to target appropriate SpO2	Max 2 L/kg/min & adjust O2 flow to target appropriate SpO2
Max flow rate	16L/min	25L/min	Small - 50 L/min Medium - 60L/min

Criteria to discuss with Cardiology Clinician (Ext 84440) & PICU Clinician (Ext 84725):

- Initiation of HiFlo in a cardiac patient on Ward 1e
- Tachypnoeic, persistent or worsening respiratory effort despite HiFlo
- Tachycardia, persistent or worsening despite HiFlo
- Sustained elevated respiratory component of PEWS score
- Clinical concern
- NB The Cardiac Ward or Hospital at Night teams will care for all other aspects of these patients care as required

Criteria to admit to PICU:

- As advised by PICU clinician
- Hi-Flo>2l/kg/min
- Clinical concern

(NB please ensure Duty Cardiology Consultant is informed of PICU admission)

Authors: D Nelson, L Bell, B Smith, M Davidson DCochran
Version 1.4
June 2024

Cardiac Ward guide for weaning Hi-Flo using AIRVO™

NHS Greater Glasgow and Clyde

Cardiac Ward guide for weaning Hi-Flo using Airvo

Is the cardiac patient clinically stable for weaning of Hi-Flo therapy?

Ensure:

- Observations stable
- No increased work of breathing
- Agreed on senior medical review to be suitable for Hi-Flo weaning
- NB if patient has balanced circulation then FiO2 & SpO2 targets will be adjusted

Monitoring

- Continuous saturation
- Hourly PEWS when weaning
- Minimum 4 hourly blood pressure & MEDICAL review twice daily

Weaning Hi-Flo Step 1 – wean oxygen

- NB if patient has balanced circulation then FiO2 & SpO2 targets will be adjusted
- If patient on less than 40% FiO2 then progress to step 2
- Review patient clinically & evaluate changes with each wean of oxygen

Weaning Hi-Flo Step 2 – wean flow

- Wean flow by 1l/min daily until a flow of 4 LPM is achieved with appropriate SpO2 in the target range and no increase in respiratory effort
- If weaning flow do not alter feeding, if feeding needs to be altered pause weaning of HiFlo
- Review patient clinically & evaluate changes with each wean of flow
- Any increase work of breathing noted, wean flow alternate days

Weaning Hi-Flo Step 3 – convert to nasal cannula oxygen

- Once oxygen of 40% or less and Flow of 4 LPM is achieved and patient stable for >24 hours convert to nasal cannula oxygen or air/oxygen blender at 3 LPM titrating FiO2 to target SpO2
- Review patient clinically & evaluate changes

Remember:

With any deterioration in the patients condition reverse previous step re-evaluate and inform medical staff.

Authors: D Nelson, L Bell, B Smith, M Davidson D Cochran
Version 1.2
Sept 2024

Medical Acute Receiving Unit (ARU) Guide for Hi-Flo (AIRVO™) Use in Bronchiolitis

INDICATIONS

- Clinical diagnosis of moderate / severe bronchiolitis with:
  - Increased respiratory effort
  - SpO2 <90% on 2L nasal cannula O2
  - ≥3kg
- Consider:
  - continuous feeds or
  - Fasting with NG on free drainage

MONITORING

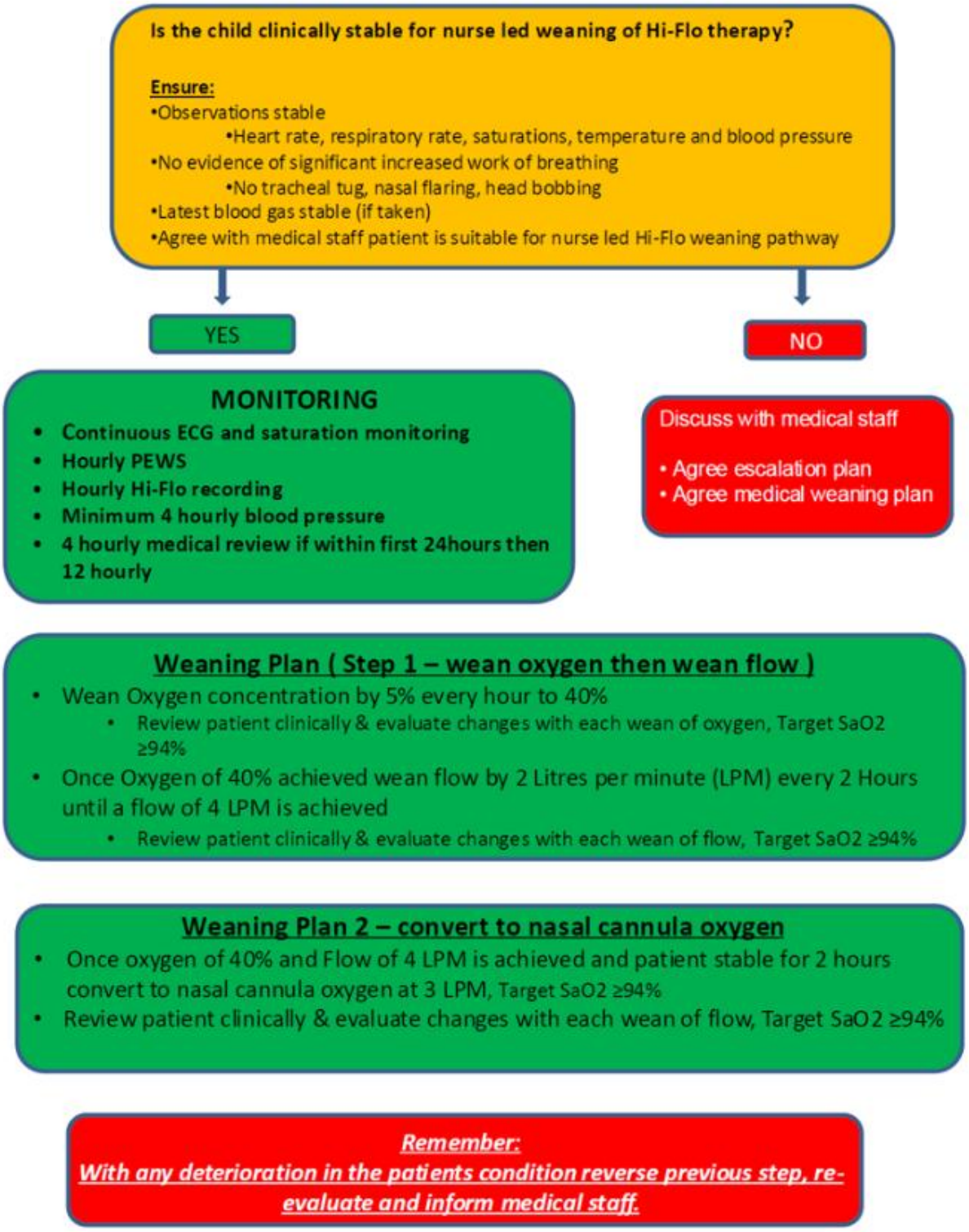
- Continuous ECG and saturation monitoring
- Hourly PEWS
- Hourly Hi-Flo recordings
- Minimum 4 hourly blood pressure
- 4 hourly medical review for first 24 hours then 12 hourly

CONTRA-INDICATIONS for Ward *Weight* HiFlo:

- < 3kg (lower weight can be discussed with on-call paediatrician especially if established in PICU and transferring to ward)
- Trauma / Surgery to nasopharynx or base of skull
- Congenital abnormality of nasopharynx e.g. choanal atresia
- Major congenital cardic defect (discuss patients with minor cardiac defects with on-call paediatric consultant)

Hi-Flo initiation	From 3kg to 8kg	8kg – 15kg
HiFlo device	AIRVO	AIRVO
Tubing Size	Junior	Junior
Cannula Size (<50% nare)	Infant (Purple)	Paediatric (Green)
Humidifier Temp.	34°C	34°C
Initial Settings	2 L/kg/min & adjust O2 flow to SpO2 ≥ 94%	2 L/kg/min & adjust O2 flow to SpO2 ≥ 94%
Max Flow Rate	20L/min	25L/min
Criteria to discuss with PICU (PICU Crash Fellow 84725) ▪Oxygen saturations (SpO2) <94% despite Hi-Flo 2L/kg/min & FiO2 100% ▪Tachypnoeic, persistent or worsening respiratory effort despite HiFlo +/- increased WOB ▪Tachycardia, persistent or worsening despite HiFlo ▪Sustained elevated respiratory component of PEWS score ▪Clinical concern		
Criteria to admit to PICU	Hi-Flo >2 L/kg/min, nCPAP or clinical concern Call PICU Crash fellow (Ext 84725) for urgent PICU review	

Medical Acute Receiving Unit (ARU) Guide for Nurse-Led Weaning of Hi-Flo (AIRVO™) in Bronchiolitis



AIRVO™2 and AIRVO™3 – Set up and Usage

For standard Operating Procedures in AIRVO™2 and AIRVO™3 please see table below for specific location of instructions for each task.

AIRVO™2 - User Manual AIRVO-manual-uk-us-and-az-ui-185045495-h-15thmarch22.pdf	
Setting up AIRVO™2	Section A4-5
Using AIRVO™2	Section A6-9
Troubleshooting	Section A10-11
Reprocessing (Disinfecting, replacing filters)	Section A12-14

AIRVO™3 – User Manual AIRVO-3-ui-620523-d2.pdf	
Setting up AIRVO™3	Section 4
Using AIRVO™3	Section 5
Monitoring Data	Section 6
Troubleshooting	Section 7
Reprocessing (Disinfecting, replacing filters)	Section 8

Instructional Videos – AIRVO™2

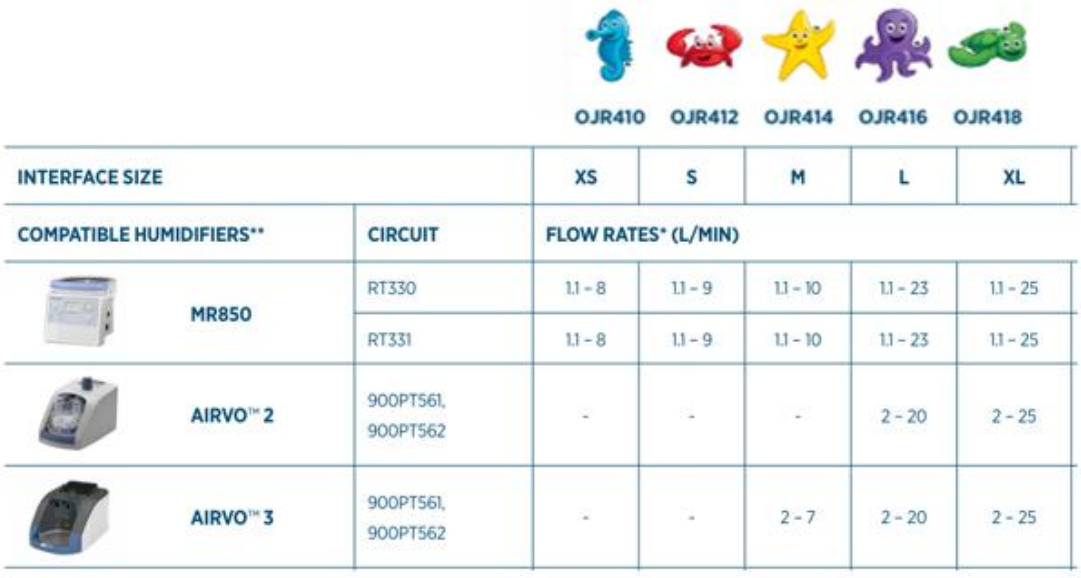
An Introduction to the F&P AIRVO™2, for Optiflow Nasal High Flow therapy	An Introduction to the F&P AIRVO™ 2, for Optiflow™ Nasal High Flow therapy - YouTube
How to set-up your F&P AIRVO™2 for Optiflow Nasal High Flow therapy	How to set-up your F&P AIRVO™ 2 for Optiflow™ Nasal High Flow therapy
How to use the F&P AIRVO™2 for Optiflow Nasal High Flow therapy	How to use the F&P AIRVO™ 2 for Optiflow™ Nasal High Flow therapy
How to Clean and Disinfect the AIRVO™2 (reprocessing between patients)	How to Clean and Disinfect the F&P AIRVO™ 2 (reprocessing between patients)

Instructional Videos – AIRVO™3

F&P AIRVO™3 Optiflow High Flow Therapy Setup	F&P AIRVO 3 Optiflow High Flow Therapy Setup
F&P AIRVO™3 Optiflow High Flow Therapy Use	F&P AIRVO 3 Optiflow High Flow Therapy Use
F&P AIRVO™3 Reprocessing – Red Disinfection Kit	F&P AIRVO 3 Reprocessing - Red Disinfection Kit

Nasal Cannula Interface Sizing Guide and Appropriate Flow Guide

Junior Interfaces



		OJR410	OJR412	OJR414	OJR416	OJR418
INTERFACE SIZE		XS	S	M	L	XL
COMPATIBLE HUMIDIFIERS**	CIRCUIT	FLOW RATES* (L/MIN)				
MR850	RT330	1.1 - 8	1.1 - 9	1.1 - 10	1.1 - 23	1.1 - 25
	RT331	1.1 - 8	1.1 - 9	1.1 - 10	1.1 - 23	1.1 - 25
AIRVO™ 2	900PT561, 900PT562	-	-	-	2 - 20	2 - 25
AIRVO™ 3	900PT561, 900PT562	-	-	2 - 7	2 - 20	2 - 25

(Inserted from optiflow-junior-2-interface-spec-sheet-pm-627621.pdf)

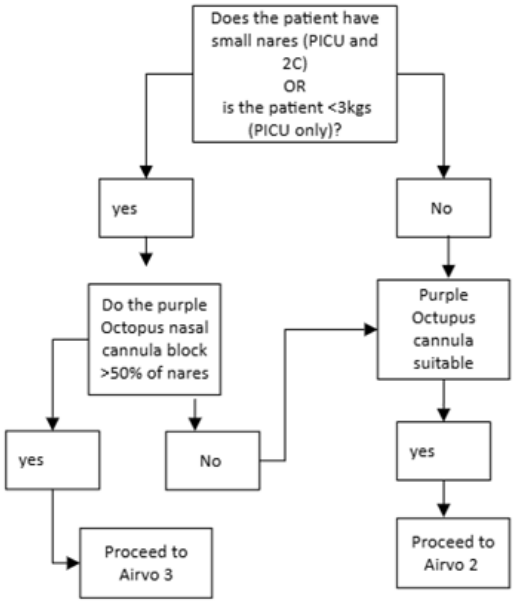
As you can see from the table above AIRVO 2™ is not licenced for the use of the Yellow OJR414 (Starfish) Interface, which is only licenced for use in the AIRVO 3™. As a result, AIRVO 3™ should be prioritised for patients weighing approx. 3-5kg and/or have nares too small to comfortably wear L nasal cannula interface. To select appropriate sized nasal cannula for your patient please see optiflow-junior-2-inservice-guide-en-sui-609691.pdf.

AIRVO™3 Patient Pathway for RHC Ward 2C and PICU

Patient assessment – inclusion criteria	**Patient assessment- exclusion criteria**
-Small nares or <3kgs (PICU only) -Requirement for heated high flow therapy -Correct interface appropriate to • Infants weight -OJR414 Nasal Cannula – ***Medium (starfish)*** (AIRVO™3 Only) • <50% occupancy of infants nare- OJR414 Nasal Cannula – ***Medium (starfish)*** (AIRVO™3 Only)	-Weight over 5kgs -Immediate need for CPAP or escalation to intubation -Upper airway obstruction requiring alternative respiratory support -Facial or nasal abnormalities/surgery preventing nasal cannula fit

Set up AIRVO 3™ and commence therapy as documented in the AIRVO™ SOP and as directed by clinicians in the appropriate clinical area

AIRVO™3 Decision Tree for RHC Ward 2C and PICU



Patients greater than 10 years old and those who have had an upper airway bypassed using a tracheostomy interface should be nursed in adult mode of AIRVO™2 - please refer to Appendix 1 (High Flow and the Adult Patient).

Oxygen Supply for AIRVO™3

To standardise oxygen supply to both AIRVO™ machines, it is recommended that the Schrader valve is inserted into the wall from the AIRVO™3 machine as seen in the picture as opposed to connecting oxygen tubing from the low flow oxygen points to the low flow meter. Target FiO2 can then be manually adjusted using the touchscreen as shown.



Patient Care whilst on AIRVO™2 or AIRVO™3

Nursing - Patient Ratio

Staff to patient ratio should be based on the overall assessment of the patient. The patient's PEWS (Paediatric Early Warning Score) should be utilised in addition to clinical assessment when determining ratio need. Patient ratios should be adjusted depending on the outcome of assessment. Flexibility in staffing should be possible due to the potential rapid change in patient dependency

Acuity	Low risk/long-term use of High Flow therapy	Medium risk	High risk
Descriptor	Actively weaning high flow or established on high flow as a long-term therapy Mild or no respiratory distress	Acute phase, some stability established but not able to wean FiO_2. Moderate respiratory distress	Acute initiation phase Severe respiratory distress
Nurse : patient ratio	1:4	1:2 or 3	1:1

Monitoring

Minimum required monitoring includes: hourly assessment of:

- Continuous oxygen saturation,
- Continuous ECG monitoring,
- Hourly PEWS assessment - respiration rate, effort and air entries (PICU only) unless on stable patient where frequency will follow routine PEWS assessment ("blow-and-grow")

Observation frequency may be reduced in clinically stable patients following consultant review and agreement. Staff should always be prepared to support with bag-mask ventilation in emergency.

Nutrition

Many patients require 'artificial feeding' (e.g. via NG/NJ/PEG) when on nasal high flow therapy.

Gastric de-venting should be performed every 2-4 hours for minimalised abdominal distention. Monitoring of child's work of breathing throughout feeding is essential.

IV fluids may be required in some circumstances, e.g. unstable patients with escalating support. The need for IV fluids should be re-assessed regularly.

Stable infants and children may be considered for oral feeding following consultant review and completion of an individual risk assessment.
Prior to oral feeding, the nurse should discuss benefits and disadvantages of oral feeding the patient on AIRVO™2 with NIC and medical staff.

Nebulised Therapy whilst receiving High-Flow

Nebulised medicines may need to be delivered during High-Flow therapy via AIRVO™2.

To use external nebuliser systems, High- Flow should be ceased whilst the nebulised medicines are given. This is undertaken by 2 options:

1. Removing the nasal cannula interface from the patients' nostrils during nebuliser, in which case the AIRVO™2 unit does not need to stop during nebulisation.
2. Turning off the AIRVO™2 and oxygen during the nebuliser administration, use wall oxygen as the driver for the nebuliser.

Using AEROGEN® adapter nebuliser kits:

Adapted circuits for use with AEROGEN® nebulisers are available (900PT562). The adapted kit allows the integration of AEROGEN Solo® nebuliser set. The AEROGEN® can stay in place for up to 28 days. The system is however only licensed for use with salbutamol and ipratropium and is therefore reserved for use in asthmatic patients.

High-Flow Therapy via Tracheostomy

If using AIRVO™2 with a tracheostomy, the machine must be set to adult mode and target temperature of 37° unless not tolerated by patients. To be used in conjunction with tracheostomy interface and tracheostomy care guidance.

It is essential to ensure the expiration valve on the tracheostomy direct connector interface is always clear of obstruction.

Please see optiflow-trach-interface-junior-sui-625321.pdf for tracheostomy interface fitting guide

Transferring/Moving a Patient on High-Flow

Patients cannot be transferred to other clinical areas on the 'MR 850' High-Flow machine or the standard AIRVO™2 device and although the AIRVO™3 has a limited battery life without mains electrical supply the same guidance below applies.

Patients receiving HFNC therapy should only be transferred when clinically necessary. During transfer, patients using MR850 or AIRVO™2 should receive appropriate non-humidified oxygen therapy unless transfer to the AIRVO™2 UPS machine is utilised (please see below).
AIRVO™3 will allow care to continue for up to a maximum of 40minutes without mains supply.

AIRVO 2™ UPS for Transfer

There are AIRVO™2 UPS (Uninterrupted Power Supply) units available for transferring patients within and out with the critical care unit in **essential** transfers only. The UPS units have a battery expectancy of no more than twenty minutes and should therefore be plugged into an electrical outlet whenever possible.

Using the AIRVO™2 UPS

1. The UPS component of these circuits needs to be turned on in order that the AIRVO™ draws mains electrical supply. Press and hold the power button on the UPS to activate
2. Set up the AIRVO™ as documented above
3. Ensure settings match desired prescription and allow machine to calibrate
4. Connect patient to AIRVO™ UPS machine by disconnecting prongs and attaching to new equipment set up on UPS machine
5. When stable, disconnect the AIRVO™ UPS from mains supply
6. The UPS battery will provide twenty minutes electrical supply
7. Monitor the "battery indicator" light. This should remain green throughout transfer, if it turns yellow, the device should be plugged into mains supply without delay to ensure continuing operation
8. On return to the clinical area, reconnect patient to a static device to allow the UPS to be recharged.

Charging the AIRVO™2 UPS

1. After each use, the UPS battery requires 12 hours to fully recharge
2. The device should be turned off and left to charge with a sign to indicate ready time

Any issues with the UPS AIRVO™ should result in its removal from use to allow investigation to occur

Maintenance Information

Reprocessing AIRVO™2
Disinfecting and replacing filters - section A12-14 - AIRVO-manual-uk-us-and-az-ui-185045495-h-15thmarch22.pdf

Reprocessing AIRVO™3
Disinfecting and replacing filters – section 8 - AIRVO-3-ui-620523-d2.pdf

The maintenance table below is applicable to both AIRVO™2 and AIRVO™3

Stock	Action	How often	Rationale
Nasal cannula/patient adaptor	Renew	Every 7 days	As per manufacturer guidance; prevent cross infection
Heated breathing tube and chamber kit	Renew	Every 14 days	
Air Filters	Renew	Every 1,000 hours or every 3 months	
Unit Cleaning	Run Disinfection cycle	After each patient	
Unit Up-keep	Fully Inspect and Damp dust	Prior to use and Daily when in use	

Documentation

Patients in PICU

Patients being treated on AIRVO™ within PICU should have their respiratory support charted on MetaVision. Patients should have settings recorded hourly and observations as per 'Section 9 - Monitoring' continuously, or at least 2-4 hourly.

Settings that are required every hour are:

1. M-FiO2,
2. Gas Flow L/min,
3. Airway temp.

Also required are:

1. Airway type (AIRVO™ + Nasal Cannula/Tracheostomy)
2. Ventilation mode (Spontaneous + Paediatric/Adult)

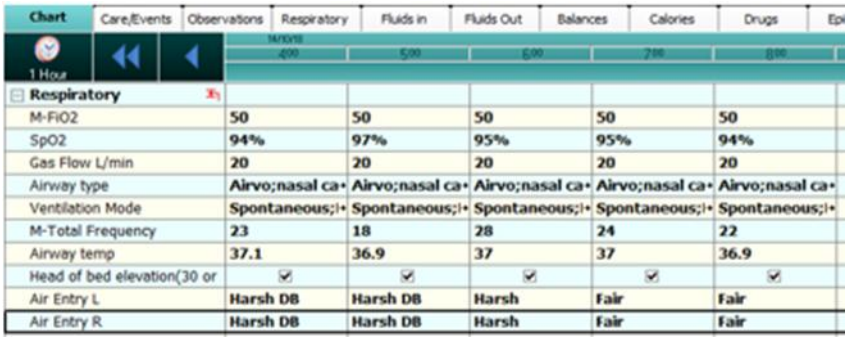


Chart	Care/Events	Observations	Respiratory	Fluids in	Fluids Out	Balances	Calories	Drugs	Epi
1 Hour			400	500	600	700	800		
Respiratory									
M-FiO2			50	50	50	50	50		
SpO2			94%	97%	95%	95%	94%		
Gas Flow L/min			20	20	20	20	20		
Airway type			Airvo;nasal ca	Airvo;nasal ca	Airvo;nasal ca	Airvo;nasal ca	Airvo;nasal ca		
Ventilation Mode			Spontaneous;I	Spontaneous;I	Spontaneous;I	Spontaneous;I	Spontaneous;I		
M-Total Frequency			23	18	28	24	22		
Airway temp			37.1	36.9	37	37	36.9		
Head of bed elevation(30 or			☑	☑	☑	☑	☑		
Air Entry L			Harsh DB	Harsh DB	Harsh	Fair	Fair		
Air Entry R			Harsh DB	Harsh DB	Harsh	Fair	Fair		

The CIS chart above represents a patient receiving AIRVO™ therapy with device settings documented: ADULT mode via Nasal Cannula with flow of 20L/min, FiO_2 of 50% and humidifier is set to 37°C.

Patients who are being transferred from PICU to down-stream wards while remaining on AIRVO™ support should, at minimum, have their AIRVO™ documentation transferred onto ward paper charts on the morning of discharge and the day of discharge be recorded on PEWS charts to allow the wards to note any relevant trends in treatment.

Ward Documentation

Ward patients receiving AIRVO™ therapy should be monitored utilising dedicated 'Hi-Flow' nursing documentation. This can be ordered via medical illustration with code numbers shown alongside images below.

HiFlo Shift Check – MI332104

Affix label
Address
CHI
Name
DOB
Hosp/ward

HiFlo Shift Check

NHS Greater Glasgow and Clyde

Date	Record HiFlo YZ number	All HiFlo equipment is intact & in good working order	HiFlo is set in the correct patient mode	HiFlo alarms are in good working order	Ensure portable O2 available in designated area	Piped suction is working	Ambu-bag & Correct size face mask available for patient	Date of next nasal cannula change	Date of next breathing tube & chamber kit change (every 14 days)	Date of next Naso-gastric tube change (Every 7 days)

*Every 7 days-short term tube. Every 28 days long term tube

mi • 332104 v1.0

Hi-Flow Hourly Check – MI332103

Affix label
Address
CHI
Name
DOB
Hosp/ward

HiFlo Hourly Check

NHS Greater Glasgow and Clyde

Today's Date:	**Nasal Cannula Size:** Naso-Gastric Tube Size: Number of Days on HiFlo:	**HiFlo PARAMETERS:** FiO2 (%): Gas Flow (litres): O2 Saturations (%): Prescriber Signature	**HiFlo WEANING PARAMETERS:** FiO2 (%): Gas Flow (litres): O2 Saturation (%): Prescriber Signature

Time	% O2	Gas Flow Rate	Humidified temp °	Water in Humidifier (Y/N)	Water in Bottle (Y/N)	Position of Nasal Cannula Checked	Condition of Nares Checked	Respiratory Effort *1	Head of Bed Elevation 30°	Suction (nasal/oral/ tracheal)	Secretion Type	Medical Review *2	Naso-Gastric Tube Aspiration (2-4hrly)	Signature
08:00														
09:00														
10:00														
11:00														
12:00														
13:00														
14:00														
15:00														
16:00														
17:00														
18:00														
19:00														

*1; Normal (N), Tracheal Tug (TT), Nasal Flaring (NF), Recession (R)

*2; 4 hourly medical review for 1st 24 hours. 12 hourly thereafter

Appendix 1: High-Flow and the Adult Patient

Thanks to Mairi Mascarenhas (Unit Educator Raigmore Hospital ICU) for help with this section

Background
Some older patients within RHC may require treatment using AIRVO™2 in Adult Mode. These patients receive the same benefits from humidified HFNC therapy as infants and smaller children.

For the purpose of this sub-guideline an "adult patient" is ANY patient who is treated with the AIRVO™ in ADULT mode

Any patient greater than 10 years old will most probably be sized to use one of the adult interfaces. These patients who require high flow, should be initially started on an AIRVO™2 with one of the "adult" sized interface. Remember that nasal cannula size is determined by the size of the patient's nostrils, therefore some "adult patients" may require a paediatric interface; if this is the case, follow the standard policy.

Equipment/Interfaces
The AIRVO™ is set up for an adult in the same way as for a paediatric patient. The same set-up pack should be used (containing: heated breathing tube, humidifier, and humidifier angle piece); a length of green oxygen tubing is still required to connect an oxygen outlet to the oxygen in port. For adult patients use the high flow oxygen meter attached to the AIRVO™, ensure the high flow tubing is plugged into a Schrader outlet on the bedside pendant.

When choosing interfaces the following are all suitable to be used when on Adult mode (remember if using an Adult interface to ensure the AIRVO™2 in Adult mode and vice versa). Please note there are no 'modes' on the AIRVO™3.

Escalation
As with any patient, the delivered therapy may not achieve the desired result. It is important to remember that even though the flows are much higher when using the adult protocol, these flows are normal for the patient's age and weight, therefore a flow of 30 or 40 litres should not cause undue alarm to nursing and medical staff. Staff should, however, be aware of increasing flows and oxygen requirements. Patients who are escalating in treatment should have a medical review, as another mode of ventilation may be required e.g. invasive or non-invasive mechanical ventilation

Appendix 2: Using an Oxygen Cylinder

There has been a Patient Safety alert due to a number of adverse events involving portable CD Oxygen cylinders. This is due to the assumption that the cylinder is already turned on and therefore when O2 tubing has been attached and a flow selected there has been no pressure or gas within the system.



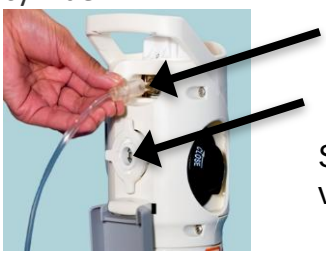
PICTURE 1



PICTURE 2



PICTURE 3

Picture 1 The valve to activate the system is hidden behind a grey panel at the side of the cylinder.	Picture 2 To reveal the valve simply pull the looped tab and remove the cap.	Picture 3 When the main control valve is revealed the black wheel can be rotated in an anti-clockwise direction to allow flow
Attach oxygen tubing from the flow meter on the pole to the Schrader Valve on the oxygen cylinder  Not used Schrader valve		

Appendix 3: Oxygenation Flow-Ratio Chart

The table below shows on the:

- *x* axis (horizontal) the total flow output programmed by the operator of the AIRVO™2 or AIRVO™3
- *y* axis (vertical) the end oxygen concentration desired by the operator of the AIRVO™

In the centre of the chart is an approximate flow to be set on the oxygen flow meter to achieve these settings.

This table is ONLY a guide, and the AIRVO™ screen should ALWAYS the observed to give accurate oxygen concentration readouts

		Total AIRVO 2™ flow set by operator											
		5	10	15	20	25	30	35	40	45	50	55	60
Desired oxygen concentration	25	1	1	1	2	2	2	2	3	3	3	3	4
	30	1	2	2	3	3	4	4	5	6	6	7	7
	35	1	2	3	4	5	6	7	8	8	9	10	11
	40	2	3	4	5	7	8	9	10	11	13	14	15
	45	2	4	5	7	8	10	11	13	14	16	17	19
	50	2	4	6	8	10	12	13	15	17	19	21	23
	55	3	5	7	9	11	13	16	18	20	22	24	26
	60	3	5	8	10	13	15	18	20	23	25	28	30
	65	3	6	9	12	14	17	20	23	26	28	31	34
	70	4	7	10	13	16	19	22	25	28	32	35	38
	75	4	7	11	14	18	21	24	28	31	33	38	42
	80	4	8	12	15	19	23	27	30	34	38	42	45
	85	5	9	13	17	21	25	29	33	37	41	45	49
	90	5	9	14	18	22	27	31	35	40	44	49	53
	95	5	10	15	19	24	29	33	38	43	47	52	57

Note:

At high flows the highest oxygen requirements may not be achievable, for example at 60 litres of flow the maximum achievable oxygen concentration (with flow meter open fully) is approximately 75%. In these cases consider escalating respiratory support!

Appendix 4: Equipment stock list for ordering

AIRVO™2 and AIRVO™3 – STANDARD USE		
	Tube and Chamber Kit	Interfaces
Junior AIRVO™2	900PT561 - Heated breathing tube with auto-fill chamber and adapter kit (Box of 10)	OJR416 Nasal Cannula – Large OJR418 Nasal Cannula – Paediatric WJR112 Wigglepads (All packets of 20)
AIRVO™3		OJR414 Nasal Cannula – Medium (AIRVO 3™ Only) OJR416 Nasal Cannula – Large OJR418 Nasal Cannula – Paediatric WJR112 Wigglepads (All packets of 20)
Adult		OPT942 Nasal Cannula – Small OPT944 Nasal Cannula – Medium OPT946 Nasal Cannula – Large OPT970 Tracheostomy – Direct Connection (All boxes of 20)
Miscellaneous AIRVO™2	900PT913 - Air Filters (Packet of 2) 900PT600 - Disinfection Kit (Packet of 1) 900PT601 - Disinfection filter (Packet of 2) 900PT603 - Clean Storage Cover (Packet of 20)	Kit 900PT600 contains: manual disinfection tube with disinfection filter/cap; two cleaning sponges; and three storage covers
Miscellaneous AIRVO™3	900PT602 – Air Filter (Pack of 2) 900PT602 – Cleaning Sponge Kits 900PT933 – Outlet Elbow 900PT600 – Disinfection Kit	

A variance of the standard Tube and Chamber Kit (900PT561) (above) is available. The varied Kit (900PT562) is adapted to allow the integration of an Aerogen Solo® nebuliser set. This adaption of the AIRVO™ device is currently licensed only for use with Salbutamol and Ipratropium and therefore is reserved for use in asthmatic patients.

THIS TUBING IS NOT STOCKED AS STANDARD IN PICU

Additional Stock /Equipment Items
Green Oxygen tubing
Water for Inhalation Carefusion 2D0735X

Contacts for Support and Training in Royal Hospital for Children

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- CNE Jenna Samborek
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Ward Contacts/trainers:
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Version Control

AIRVO™2 Standard Operating Procedure	
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Lead Authors:	John Thomson, Lyndsay Burns
Specialist advice	Mairi Mascarenhas for adolescent input
Previous approval Group: Local specialists	Dr Mark Davidson; Dr Neil Spenceley; Jenna Hills; Stephen McGloin; Danilo Cajilis; Janette Grady; Darren Eskdale.
Current Approval Group	PICU Guideline Group, Medical Paediatrics, Cardiology ANPs
Changes to clinical content of this version:	- Widening to hospital policy therefore updated for specific areas and needs - Updated product codes - Updated wording - Updated in-text links - Addition of AIRVO 3™ (arrived May 2026)

Related Documents

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